INCH-POUND

MIL-DTL-83413/4E
w/AMENDMENT 2
29 May 2019
SUPERSEDING
w/AMENDMENT 1
21 December 2018

DETAIL SPECIFICATION SHEET

CONNECTORS AND ASSEMBLIES, ELECTRICAL, AIRCRAFT GROUNDING: PLUGS, FOR TYPES I AND II GROUNDING ASSEMBLIES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83413.

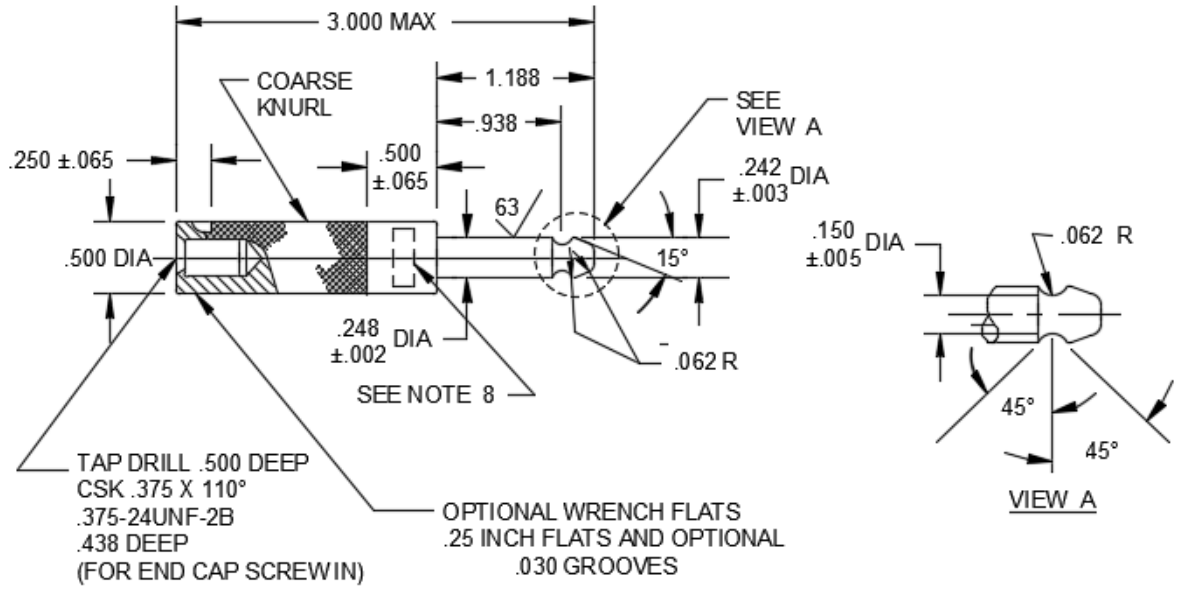


FIGURE 1. Plug, for types I and II grounding assemblies.

A nominal 7/16-inch hexagonal shaped plug body may be used in lieu of the cylindrical shape shown on figure 1. If the hexagonal shaped body is used, knurling is not required and the .250 and .500 dimensions for positioning the knurling are not applicable.

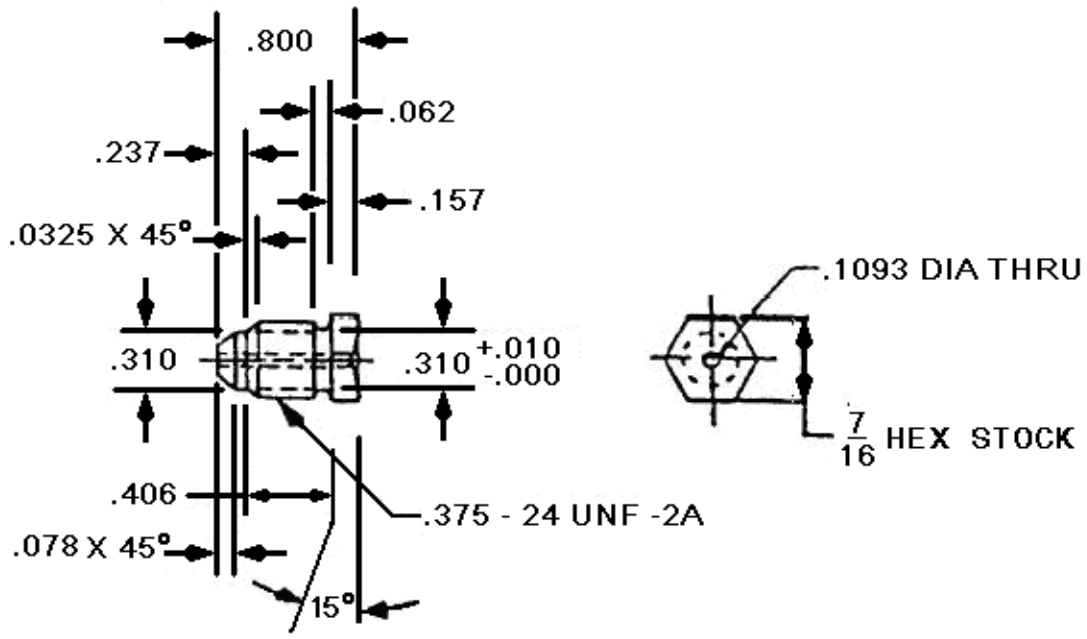
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:01Z
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm
.062	1.57	.438	11.13
.150	3.81	.500	12.70
.242	6.15	.938	23.83
.248	6.30	1.188	30.18
.250	6.35	3.000	76.20
.375	9.53		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .016 (.41mm) for decimals and ± 0.5° for angles.
4. This plug will be used with a receptacle having a minimum bore diameter of .251 (6.38 mm) and a maximum bore diameter of .253 (6.43 mm) in accordance with ISO 46 and NATO STANAG 3632. Plugs must mate with receptacles within these dimensions to achieve mandatory international interoperability.
5. Break all sharp edges .010 - .020 (.25 mm – 51 mm).
6. The M83413/4-3 plug supersedes plugs with both cylindrical and hexagonal body shapes and various tip dimensions and is functionally interchangeable, although mating forces may vary slightly.
7. Optional hexagonal body shape allowed, see requirements.
8. Part or Identifying Number (PIN). In accordance with MIL-STD-1285.

FIGURE 1. Plug, for types I and II grounding assemblies – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:01Z
Check the source to verify that this is the current version before use.



Inches	mm	Inches	mm
.0325	0.83	.237	6.02
.062	1.57	.310	7.87
.078	1.98	.375	9.53
.1093	2.78	.406	10.31
.150	3.81	.800	20.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .016 (.41mm) for decimals and ± 0.5° for angles.

FIGURE 2. End cap.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:01Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.

Material: Stainless steel in accordance with SAE-AMS-S-7720, 300 series composition, condition A, passivated in accordance with SAE-AMS2700 for plugs and caps.
Dust: Not applicable.
Tensile strength: Not applicable.
Engagement: Not applicable.
Crush: Not applicable.
Durability: Durability test shall be performed in accordance with test procedure EIA-364-09. The following conditions shall apply:

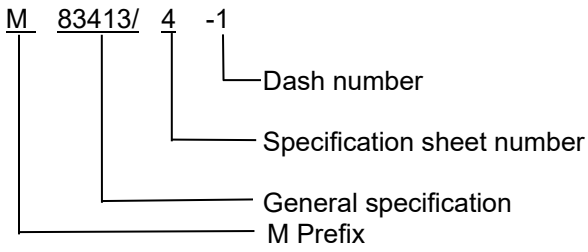
a. Test preparation: Connector shall be mated.
b. 3000 cycles shall be completed.
c. After 3000 cycles, the detent action test and the contact resistance test shall be performed. Contact resistance shall be .005 ohm maximum.

Contact resistance: .005 ohm maximum.

Mating receptacle: In accordance with MIL-DTL-83413/5 or MIL-DTL-83413/6. Use MIL-DTL-83413/6 for all qualification and quality conformance tests.

PIN: Apply permanently by stamping, etching, embossing or other means approved by the qualifying activity. Dash number marking is optional on PIN M83413/4-1 and -3. PIN is not required to be marked on M83413/4-2 end caps.

Example of PIN: M83413/4-1



Supersession data: See table I.

TABLE I. Supersession data.

PIN	Description	Superseded PIN
M83413/4-1	M83413/4-3 plug with M83413/4-2 end cap attached	MS3493-3 MS3493-5
M83413/4-2	7/16 inch end cap	MS3493-2
M83413/4-3	Knurled cylindrical plug or unknurled hexagonal plug	MS3493-1 MS3493-4

M83413/4-1 shall be supplied with the end caps screwed in the mating plug end.

A nominal 7/16-inch hexagonal shaped plug body may be used in lieu of the cylindrical shape shown on figure 1. If the hexagonal shaped body is used, knurling is not required and the .250 and .500 dimensions for positioning the knurling are not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:01Z
Check the source to verify that this is the current version before use.

Marking: Connectors shall be permanently and legibly marked in accordance with method I of MIL-STD-1285 with the PIN, and the JAN certification mark. Product packaging labels shall be marked with the manufacturer's name or trademark, cage code and date code.

JAN and J marking: The United States Government has adopted, and is exercising legitimate control over the certification marks “JAN” and “J”, respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of specifications. Accordingly, items acquired to, and meeting all of the criteria specified herein and in applicable specifications shall bear the certification mark “JAN” except that items too small to bear the certification mark “JAN” shall bear the letter “J”. The “JAN” or “J” shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the “JAN” or “J” may be located on the first line above or below the part number. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable specifications shall not bear “JAN” or “J”. In the event an item fails to meet the requirements of this specification and the applicable specification sheets, the manufacturer shall remove completely the military part number and the “JAN” or the “J” from the sample tested and also from all items represented by the sample. The “JAN” and “J” certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration Number 504,860 for the certification mark “JAN” and Registration Number 2,577,735 for the certification mark “J”.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:01Z
Check the source to verify that this is the current version before use.

Qualification: Required. The activity responsible for qualification and the Qualification Products List (QPL) for this specification sheet is the DLA Land & Maritime- VQ, 3990 East Broad Street, Columbus, Ohio 43218-3990.

To retain qualification, the qualified manufacturer shall periodically forward reports to the Qualifying Activity in accordance with MIL-DTL-83413. Qualification inspection shall consist of the inspections specified in table II.

TABLE II. Qualification inspection, connectors and jumper assemblies. 1/

Inspection	Plug	Receptacle	Clamp connector	Jumper assembly	Requirement paragraph	Test Paragraph
Group I						
Visual and mechanical Inspection	X				3.1, 3.3, 3.4 3.6, 3.8	4.7.1
Detent action	X				3.5.1	4.7.2
Handgrip force					3.5.2	4.7.3
Side loading					3.5.3	4.7.4
Torque					3.5.4	4.7.5
Engagement					3.5.5	4.7.6
Contact resistance	X				3.5.6.1	4.7.7.1
Clamp connector resistance					3.5.6.2	4.7.7.2
Assembly resistance					3.5.6.3	4.7.7.3
Group II						
Cable attachment					3.5.7	4.7.8
Clamp release force (initial)					3.5.8.1	4.7.9
Clamp release force (after conditioning)					3.5.8.2	4.7.9
Clamping life					3.5.9	4.7.10
Vibration					3.5.10	4.7.11
Durability	X				3.5.11	4.7.12
Flexibility					3.5.13	4.7.14
Tensile strength					3.5.12	4.7.13
Crush					3.5.14	4..7.15
Group III						
Humidity	X				3.5.15	4.7.16
Temperature cycling	X				3.5.16	4.7.17
Drop					3.5.17	4.7.18
Altitude – low temperature					3.5.18	4.7.19
Dust					3.5.19	4.7.20
Fluid immersion					3.5.20	4.7.21
Salt spray	X				3.5.22	4.7.23
Group IV						
Lightning current					3.5.23	4.7.24

1/ - X indicates that a test is applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:01Z
Check the source to verify that this is the current version before use.

Certain provisions of this specification sheet are the subject of International Standardization Agreement NATO STANAG 3632 and ISO 46. When amendment, revision or cancellation of this specification sheet is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international agreement standardization channels, including departmental standardization offices to change the agreement or make other appropriate accommodations.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-83413, this document references the following:

MIL-DLT-83413/5
MIL-DTL-83413/6
MIL-STD-1285
SAE-AMS2700
SAE-AMS-S-7720
STANAG 3632
ISO 46
EIA-364-09

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy – AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army - AR, CR, CR4, MI
Navy - MC
Air Force – 02

(Project: 5935–2019–062)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:01Z
Check the source to verify that this is the current version before use.